

Architectural Optimization of 'La Ghigliottina' Solver

University of Bari Aldo Moro

1. The NLP Task: 'La Ghigliottina'

- **Context and Game Rules**

- A lexical association problem requiring lateral thinking.
- Input: 5 seemingly unrelated clue words (hints).
- Output: 1 target word sharing a semantic, syntactic, or cultural link.

- **Scientific Goal**

- Overcome the 'Knowledge Bottleneck' of deterministic systems (pure logic).
- Mitigate the 'Entropic Noise' of massive free-text ingestion (pure statistics).

2. Dual Data Pool Configuration

- **Symbolic Knowledge (Absolute Truths)**
 - Local, highly qualified domain dictionaries.
 - ~36,000 rows of exact lexical intersections (proverbs, multi-word expressions).
- **Distributional Knowledge (Statistical Semantics)**
 - >11 GB of raw unstructured text in asymmetrical configuration.
 - Paisà Corpus: ~2.1M sentences.
 - Wikipedia (Italian): ~1.58M lines.
 - OpenSubtitles: ~37.7M utterances.

3. SOTA Retrieve-and-Rerank Pipeline

- **Phase 1: Deterministic Retrieval (Symbolic Filter)**
 - Lexical Knowledge Graph (LKG) performs Path-Based Reasoning in $O(1)$ time.
 - Extracts an ultra-high recall candidate pool (Top-50).
- **Phase 2: Distributional Feature Injection**
 - Sparse Co-occurrence Engine computes NPMI metric from texts.
- **Phase 3: Supervised Reranking (XGBoost)**
 - Vector space defined in $\mathbb{R}^8$.
 - Dynamically balances topological rigidity with semantic statistics.

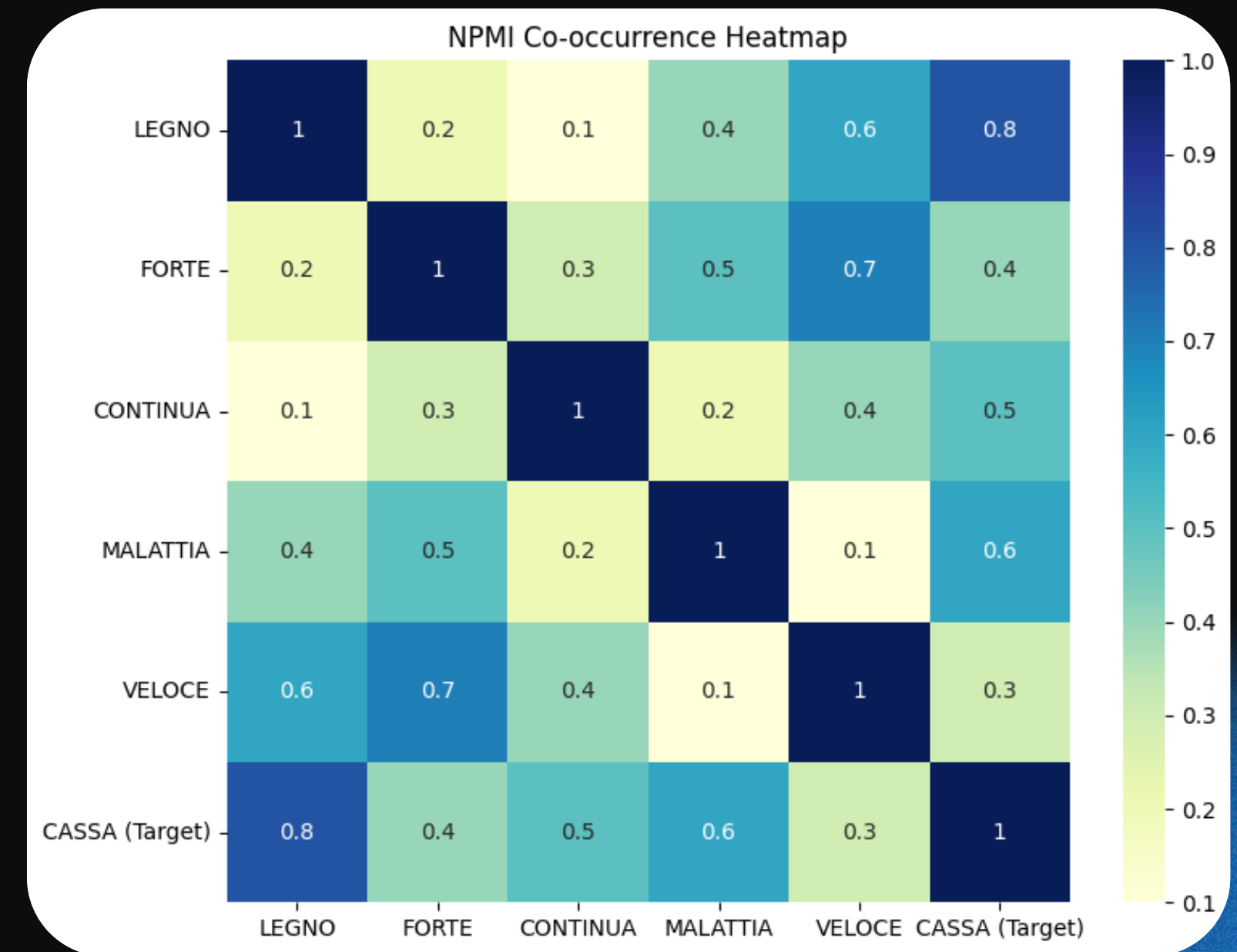
4. Operational Logic & Optimizations

- **Streamed Data Ingestion**

- Lazy evaluation via Python generators prevents system crashes.
- Set-theoretic normalization ensures $O(1)$ lookup during stopwords removal.

- **High-Dimensional Matrix Compression**

- A dense matrix for a 1.6M vocabulary requires ~10.24 TB of RAM.
- Utilization of Compressed Sparse Row (CSR) matrices offloads tensors.
- Mathematical pruning ($f < 2$) compresses the footprint to a few Gigabytes.



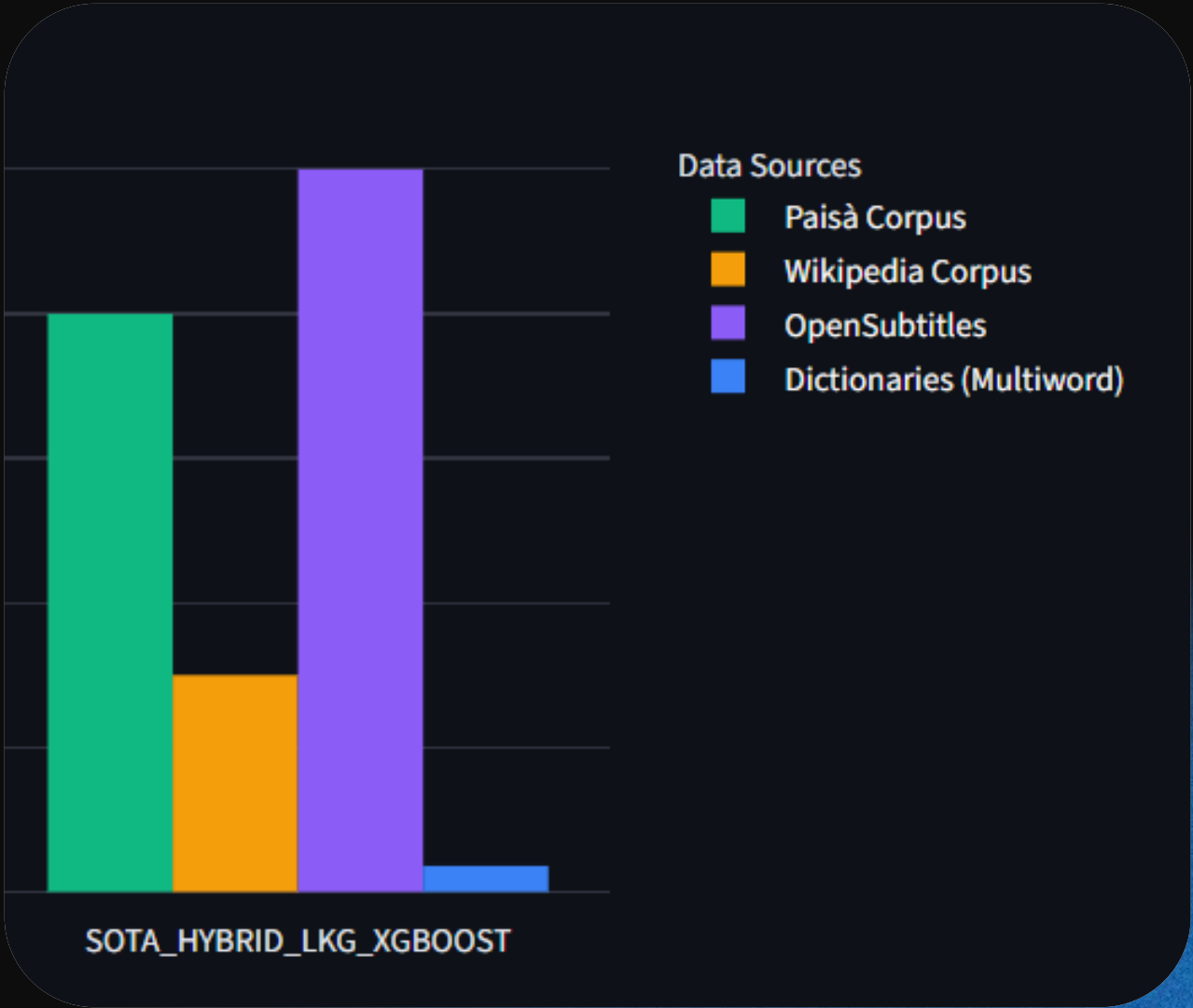
5. Critical Methodology: Data Leakage

- **The 'train.json' Hold-Out Split (80/20)**
 - 80% - Topological Injection:
 - Populates the Graph with explicit television history labels.
 - 20% - XGBoost Vectorization:
 - Undergoes Retrieval for binarized training of the decision trees.
- **Why this is mandatory:**
 - Prevents the model from artificially saturating Information Gain on the historical feature, forcing it to learn pure distributional statistics.

6. Empirical Results (Blind Test Set)

Acc @1	Acc @5	MRR
33.00%	49.00%	0.3987

- **Baseline Performances**
 - Symbolic Only: 21.00% (Accuracy @1) / 42.00% (Accuracy @5) - **Dictionary**
 - Distributional Only: ~19.00-21.00% (Accuracy @1) – **Corpora**
- **Ensemble Hybrid LTR (LKG + XGBoost) – Dictionary + Corpora**
 - Accuracy @1 (Exact Match): 33.00%
 - Accuracy @5 (Top-5 Window): 49.00%
 - Mean Reciprocal Rank (MRR): 0.3987
 - Inference Time: < 0.15s per game



SOTA_HYBRID_LKG_XGBOOST

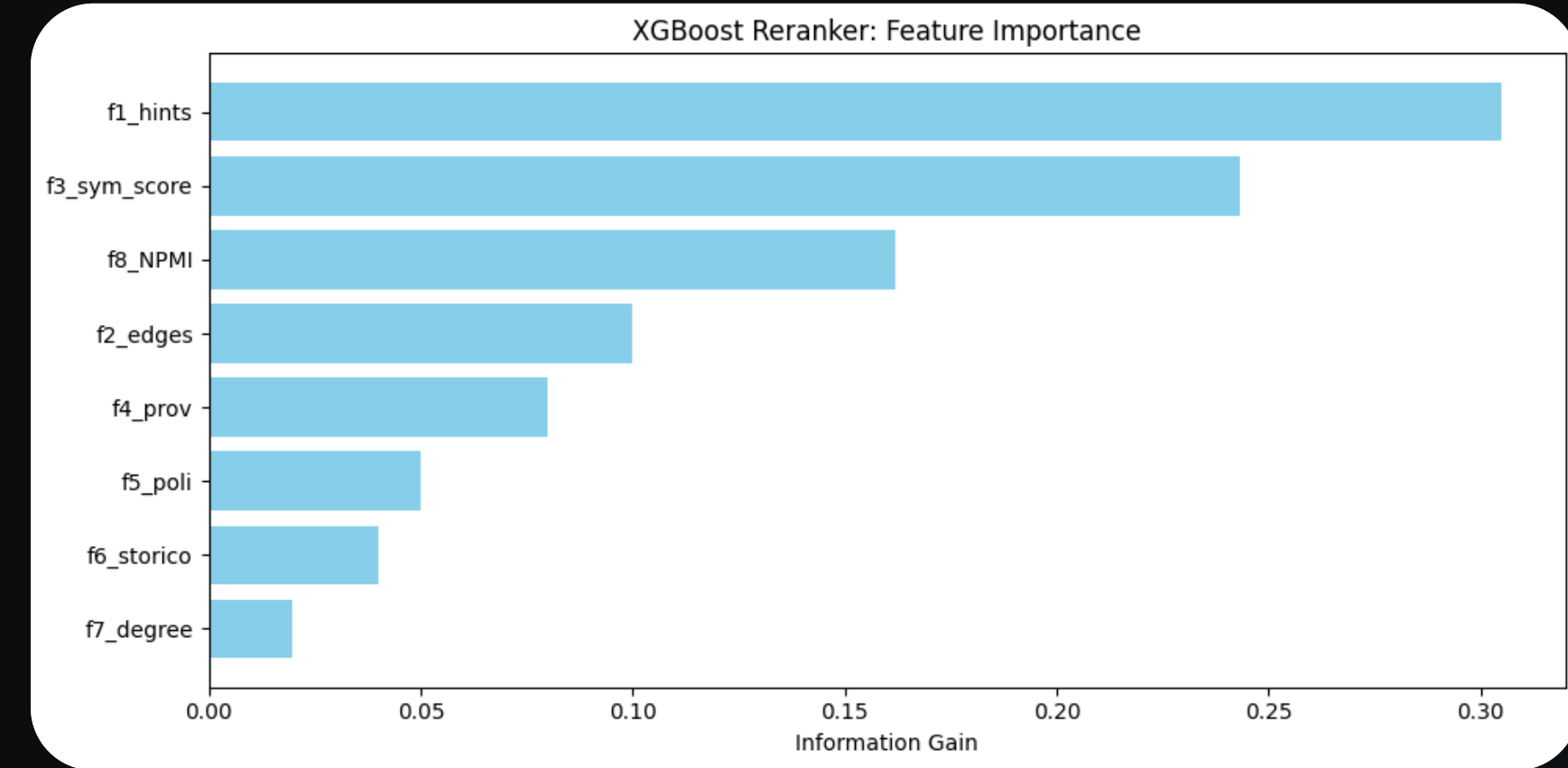
7. XGBoost Regularization Proof

- **Analysis of Information Gain**

- Demonstrates proper regularization across the decision trees.
- Distributional Statistical Feature (f_8 NPMI): ~16% decision weight.
- Topological Feature (f_1 Unique Hints): ~30% decision weight.
- Topological Feature (f_3 Symbolic Score): ~24% decision weight.

- **Conclusion**

- The model successfully learned to rely on web-based semantics when deterministic graph intersections were insufficient.



8. Explainable AI via LLMs

- **Zero-Shot Generative Reasoning**
 - Integration of asynchronous Groq APIs (LLaMA-3.3-70B).
 - Implemented Asynchronous Throttling (1.5s delay) to prevent HTTP 429 errors.
- **Automated NLG Academic Evaluation**
 - BLEU Score (Precision): ~0.11
 - ROUGE-L F1 (Recall): ~0.33
 - BERTScore F1 (Vector Semantics): ~0.76

9. XAI Metrics: Structural Divergence

Why is BERTScore high (~0.76) while BLEU is low (~0.11)?

N-Gram Limitations (BLEU & ROUGE):

String-matching heavily penalizes the LLM for using discursive, comprehensive phrasing absent from the telegraphic human ground truth.

Vector Semantics (BERTScore):

Bypasses string-matching by computing pairwise cosine similarity on contextual token embeddings.

Confirms the model successfully captured the underlying associative logic.

10. Conclusions & Limitations

- **Key Architectural Takeaways**

- Pure syntactic knowledge requires distributional regularization to disambiguate lateral thinking (+12% Accuracy @1 leap).
- XGBoost optimally balances deterministic rigor with statistical probability.

- **Current Limitations**

- High RAM dependency for loading and computing massive NPMI matrices.
- An acceptable engineering trade-off for a real-time, robust SOTA solver.